

# OBSTRUCTION RESIDUES FOR THE $3x - 1$ MAP: FULL FACTOR COMPLEXITY, POSITIVE DENSITY, AND A CONSTRUCTIVE LIFT

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**ABSTRACT.** We study a family  $\mathcal{O}_L \subseteq \{1, \dots, 2^L\}$  of residue classes modulo  $2^L$  arising naturally in the dynamics of the  $3x - 1$  map  $T_-(n) := (3n-1)/2^{v_2(3n-1)}$ . The family is defined by an algebraic criterion on a two-track parallel reduction. We prove three structural results: a constructive lift between modular levels, the existence of a positive asymptotic density  $c_W$  with rigorous lower bound  $c_W \geq 7556/65536 \approx 0.115$ , and full factor complexity  $p_W(n) = 2^n$  of the associated language of binary representations. The central technical tool is a parity lemma in the residue system, which forces the terminal data of the parallel reduction into a shift- $a$  normal form with integer-valued shift index. A finer classification by shift index yields a strict lift balance, an exact series representation of  $c_W$ , and reduces the open question of  $c_W$ 's exact value to a single quantitative bound on a sub-family of atomic obstructions. Results transfer to the classical  $3n + 1$  map via a residue-level involution  $r \mapsto 2^L - r$  that maps the  $T_-$  parallel reduction to its  $T_+$  analogue step-by-step.

## 1. INTRODUCTION

**1.1. The Collatz map and its  $3x - 1$  cousin.** Let  $\mathbb{N}_{\text{odd}}$  denote the set of positive odd integers, and let  $v_2(m)$  denote the 2-adic valuation of an integer  $m \neq 0$ . The *Collatz map*  $T_+$  takes an odd integer  $n$  to the odd part of  $3n + 1$ :

$$T_+(n) := \frac{3n + 1}{2^{v_2(3n+1)}}.$$

The Collatz conjecture, attributed to Lothar Collatz and widely studied since the 1930s, asserts that for every  $n \in \mathbb{N}_{\text{odd}}$  the iteration  $n, T_+(n), T_+^2(n), \dots$  eventually reaches the fixed point 1. Despite considerable computational evidence and a large body of partial results — early density-side analyses by Terras [Ter76] and Everett [Eve77], refined density estimates by Crandall [Cra78] and Korec [Kor94], and most recently the breakthrough of Tao [Tao22], showing that almost all trajectories attain almost bounded values — the conjecture itself remains open. We refer the reader to the

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annotated bibliographies of Lagarias [Lag10b, Lag10a] for a comprehensive overview of the literature.

The Collatz map has a natural cousin, the  $3x - 1$  map  $T_-$  acting on the same domain:

$$T_-(n) := \frac{3n - 1}{2^{v_2(3n-1)}}.$$

The two maps are structurally parallel: both perform the operation “multiply by three, add a small constant, strip the factors of two from the result”. They differ only in the sign of the additive constant. The trajectories of  $T_-$  are, however, *not* believed to converge to a single attractor: three nontrivial cycles are classically known — the fixed cycle through 1, the  $\{5, 7\}$ -cycle, and the seventeen-family cycle through  $\{17, 25, 37, 55, 65, 49, 73, \dots\}$ ; see Lagarias [Lag10a] for historical attribution. The question of whether further cycles exist remains as open as the Collatz conjecture itself. The two maps are connected through a classical identity going back to Lagarias [Lag85]: every  $T_-$ -trajectory starting from  $m$  corresponds, via the substitution  $r = 2m + 1$ , to a single  $T_+$ -step followed by a  $T_+$ -trajectory. This *Lagarias intertwining* illustrates that any structural feature of  $T_-$  has a  $T_+$  counterpart; the explicit transfer we use in Section 7, however, is different — a residue-level involution  $r \mapsto 2^L - r$  that conjugates the  $T_-$  parallel reduction into the  $T_+$  parallel reduction step-by-step (Lemma 7.2).

The present paper is concerned with a family of residue classes modulo  $2^L$  that arises naturally in the dynamics of  $T_-$ . We show that this family — which we shall call the *obstruction residues* for reasons made precise in Section 2 — has three structural properties:

- (1) a constructive lift between modular levels  $L$  and  $L + 1$ ,
- (2) positive asymptotic density in  $\{1, \dots, 2^L\}$  as  $L \rightarrow \infty$ ,
- (3) maximal combinatorial complexity of its associated language of bit strings.

The third property is in some tension with the first two: a constructive lift and positive density would, in many natural settings, force a certain rigidity of the resulting word language (one might expect a sofic shift or a primitive substitution shift). We prove that no such substitution-type characterization is possible, identifying the obstruction residues as a structurally richer family than the naive analogies would suggest.

**1.2. Parallel reductions and the  $X$ -invariant.** The construction is built on a simple observation. For an odd integer  $r$  with  $v := v_2(r - 1) \geq 1$ , write  $m := (r - 1)/2^v$ . We run  $T_-$  in parallel on  $r$  and on  $m$  modulo  $2^L$  for a fixed level  $L$ , keeping track at each step both of the current values and of their *affine relationship*: a pair  $(c_j, d_j) \in \mathbb{Q}^2$  such that  $T_-^j(m) \equiv c_j \cdot T_-^j(r) + d_j$  modulo an appropriately reduced power of two. The pair is updated by an explicit rule depending on the 2-adic valuations of the two trajectories. We denote the terminal pair  $(c_{\text{end}}, d_{\text{end}})$  and define the scalar invariant

$$X_{\text{end}}(r, L) := 3d_{\text{end}}(r, L) + c_{\text{end}}(r, L).$$

For an odd integer  $r$  with  $v_2(r - 1) \geq 1$  and a level  $L > v_2(r - 1)$ , we call  $r$  an *obstruction residue at level  $L$*  when  $X_{\text{end}}(r, L) = 1$ , and write  $\mathcal{O}_L \subseteq \{1, \dots, 2^L\}$  for the set of all such residues. Definition 2.2 in Section 2 restates this with the parallel reduction made explicit; the typical case is  $v = 1$ , i.e.  $r \equiv 3 \pmod{4}$ .

The family  $(\mathcal{O}_L)_{L \geq 5}$  is small at first —  $\mathcal{O}_5 = \{27\}$ ,  $\mathcal{O}_6 = \{19, 27, 59\}$  — but grows rapidly: direct enumeration gives

$$|\mathcal{O}_5| = 1, \quad |\mathcal{O}_6| = 3, \quad |\mathcal{O}_7| = 8, \quad |\mathcal{O}_8| = 19, \quad \dots, \quad |\mathcal{O}_{16}| = 7556,$$

and the density ratio  $|\mathcal{O}_L|/2^L$  rises monotonically. Full counts at all levels  $L = 5, \dots, 16$  together with the shift-zero / shift-nonzero and atomic-shift-nonzero decompositions are collected in Table 3 (Appendix A). Three structural questions arise:

- (Q1) Does  $r_0 \in \mathcal{O}_{L_0}$  guarantee that both  $r_0$  and  $r_0 + 2^{L_0}$  lie in  $\mathcal{O}_{L_0+1}$ ?
- (Q2) Does  $|\mathcal{O}_L|/2^L$  converge to a positive limit, and can the limit be bounded from below?
- (Q3) What is the set of binary words appearing as factors of the bit representations of residues in  $\bigcup_L \mathcal{O}_L$ ?

The three questions are not independent: a positive answer to (Q1) gives monotonicity, hence most of (Q2). A constructive form of (Q1), exhibiting arbitrary bit patterns under lift, gives (Q3). All three are answered in the affirmative by the main results; in particular the factor language  $\mathcal{F}(\mathcal{O})$  defined below turns out to be all of  $\{0, 1\}^*$ .

### 1.3. Main results.

**Theorem 1.1** (Constructive lift). *Let  $r_0 \in \mathcal{O}_{L_0}$ . Then both  $r_+ := r_0$  and  $r_- := r_0 + 2^{L_0}$  are obstruction residues at level  $L_0 + 1$ .*

The proof, given in Section 3, proceeds by case analysis on a finite invariant of the parallel reduction. (No explicit lower bound on  $L_0$  is needed: for  $L_0 \leq 4$  the family  $\mathcal{O}_{L_0}$  is empty (Remark 2.3) and the statement is vacuous; the smallest non-vacuous level is  $L_0 = 5$  with  $\mathcal{O}_5 = \{27\}$ . The proof in Section 3 only uses  $L_0 > v$ , which is automatic from  $r_0 \in \mathcal{O}_{L_0}$  by Definition 2.2.) The lift is explicit: for each  $r_0$ , the terminal data of the reduction at  $r_+$  or  $r_-$  is determined by a closed-form algebraic identity.

**Theorem 1.2** (Positive density). *The density ratio  $|\mathcal{O}_L|/2^L$  is non-decreasing in  $L$ , and the limit*

$$c_W := \lim_{L \rightarrow \infty} \frac{|\mathcal{O}_L|}{2^L}$$

*exists in  $(0, 1]$ . For every  $L_0 \geq 5$ , the rigorous lower bound  $c_W \geq |\mathcal{O}_{L_0}|/2^{L_0}$  holds; at  $L_0 = 16$  this gives  $c_W \geq 7556/65536 \approx 0.115$ , sharpening the conservative bound  $c_W \geq 3/64 \approx 0.047$  obtained from  $L_0 = 6$ .*

The bound at  $L_0 = 16$ , using only the algorithmic  $X_{\text{end}}$ -criterion of Section 2, gives the explicit value  $7556/65536 \approx 0.115$  stated above. Raising

$L_0$  further admits substantial additional improvement, but the exact value of  $c_W$  remains an open problem. We give in Section 6 a refined decomposition that exhibits  $c_W$  as a rigorously convergent series and isolates the remaining open question to a single quantitative bound on a sub-family of obstructions.

For  $r \in \mathcal{O}_L$ , write  $w_r \in \{0, 1\}^L$  for the binary representation of  $r$  as an  $L$ -bit word, least significant bit first. The *factor language* of the obstruction family is

$$\mathcal{F}(\mathcal{O}) := \{u \in \{0, 1\}^* : u \text{ is a factor of some } w_r, r \in \mathcal{O}_L \text{ for some } L\},$$

and the *factor complexity function* is  $p_W(n) := |\mathcal{F}(\mathcal{O}) \cap \{0, 1\}^n|$ .

**Theorem 1.3** (Full factor complexity). *For every  $n \geq 0$ ,*

$$p_W(n) = 2^n.$$

The proof, in Section 5, is constructive: for each binary word  $u$ , we exhibit an explicit residue  $r(u)$  in some  $\mathcal{O}_L$  having  $u$  as a factor. The construction is a direct iteration of the lift theorem.

A consequence of Theorem 1.3 is a sharp structural rigidity (Corollary 5.3): the obstruction family cannot be described as a subshift of finite type, a sofic shift, or a primitive substitution shift in the sense of Pansiot [Pan84] or Devyatov [Dev15].

**1.4. Strategy.** All three theorems rest on a single algebraic structure: the *shift- $a$  normal form* of the terminal data. We prove in Section 2 (Lemma 2.4) that whenever the parallel reduction terminates with  $X_{\text{end}} = 1$ , the terminal pair has the form

$$(c_{\text{end}}, d_{\text{end}}) = \left(4^a, \frac{1 - 4^a}{3}\right) \quad \text{for some } a \in \mathbb{Z}.$$

The integer  $a$ , which we call the *shift index*, measures the difference between the 2-adic valuations of the two parallel trajectories. Theorem 1.1 is proved by case analysis on how the shift index evolves under lift; Theorem 1.3 is the iterative consequence; Theorem 1.2 follows by monotonicity from the lift. A refined classification by shift index (Section 6) then re-expresses  $c_W$  as a convergent series and isolates the remaining open question.

The shift- $a$  normal form is, to the best of our knowledge, new; in particular, it differs from the Markov-operator setup of Wirsching [Wir98], which works with an invariant probability-density function rather than a terminal algebraic datum. The remaining analysis proceeds by elementary case analysis and combinatorial bookkeeping; the arguments do not depend on specialized Collatz machinery beyond the Lagarias intertwining.

**1.5. Outline.** Section 2 introduces the parallel reduction, defines the  $X$ -invariant rigorously, and proves the parity lemma. Section 3 contains the proof of Theorem 1.1, split into the two stop-type cases. Section 4 derives Theorem 1.2 as a direct corollary. Section 5 proves Theorem 1.3 and derives the structural rigidity corollary. Section 6 develops the classification by shift

index, proves the strict lift balance and series representation, and isolates the remaining open problem. Section 7 transfers the results to the  $3x + 1$  side via the residue bijection. Section 8 discusses open questions. Appendix A summarizes computational verifications.

## 2. THE PARALLEL REDUCTION AND THE $X$ -INVARIANT

**2.1. The basic step.** Recall that for an odd integer  $n$ , the map  $T_-$  strips the 2-adic part from  $3n - 1$ :

$$T_-(n) = \frac{3n - 1}{2^{v_2(3n-1)}}.$$

We need a version of this step that keeps track of the modulus. Fix a level  $L \geq 1$ , and consider a pair  $(\alpha, \beta)$  with  $\alpha$  a positive odd integer and  $\beta$  a power of two, interpreted as a residue  $\alpha$  modulo  $\beta$ . A single  $T_-$ -step on such a pair produces

$$(\alpha', \beta') = \left( \frac{3\alpha - 1}{2^{v_\alpha}}, \frac{3\beta}{2^{v_\alpha}} \right), \quad v_\alpha := v_2(3\alpha - 1),$$

provided  $v_\alpha < v_2(\beta)$ . If  $v_\alpha \geq v_2(\beta)$ , the modulus has been exhausted and the step is undefined; we say the reduction has *terminated*.

Since 3 is odd,  $v_2(3\beta) = v_2(\beta)$ , so  $v_2(\beta') = v_2(\beta) - v_\alpha$ : the modulus strictly decreases at each step, and the reduction always terminates after finitely many steps. Iterates of  $\beta$  are in general of the form  $3^j \cdot 2^k$  rather than literal powers of two; only the 2-adic valuation  $v_2(\beta)$  is used downstream, so we track  $\beta$  as a positive integer of known 2-adic valuation.

**2.2. The two-track setup.** Fix  $r$  a positive odd integer with  $r \neq 1$ , set  $v := v_2(r - 1)$ , and write  $m := (r - 1)/2^v$ ; then  $m$  is odd. The inequality  $v \geq 1$  is automatic for odd  $r > 1$  since  $r - 1$  is then even. (We use  $v$  as a parameter throughout; the special case  $v = 1$  — equivalently  $r \equiv 3 \pmod{4}$  — is the typical reader picture and is presented first wherever a concrete calculation helps.)

Fix a level  $L > v$ . We run two  $T_-$ -reductions in parallel:

- the *K-track*, starting from  $(a_K^{(0)}, b_K^{(0)}) := (r, 2^L)$ ,
- the *I-track*, starting from  $(a_I^{(0)}, b_I^{(0)}) := (m, 2^{L-v})$ .

The initial relation  $m = (r - 1)/2^v$  writes as  $a_I^{(0)} = 2^{-v}a_K^{(0)} - 2^{-v}$ . At each step  $j \geq 0$ , we update both tracks according to the basic step, denoting  $v_K^{(j)} := v_2(3a_K^{(j)} - 1)$  and  $v_I^{(j)} := v_2(3a_I^{(j)} - 1)$ . Cumulative valuations along the two tracks are denoted  $V_K^{(j)} := \sum_{i < j} v_K^{(i)}$  and  $V_I^{(j)} := \sum_{i < j} v_I^{(i)}$ , so that the residual 2-adic valuation of  $b_K^{(j)}$  is  $L - V_K^{(j)}$  and that of  $b_I^{(j)}$  is  $L - v - V_I^{(j)}$ . The reduction proceeds as long as both tracks can step; it terminates as soon as either condition fails. We write  $J = J(r, L)$  for the index of termination.

### 2.3. The affine relation and the $X$ -invariant.

**Lemma 2.1.** *For each  $j$  with  $0 \leq j \leq J$ , there exist  $c_j, d_j \in \mathbb{Q}$  such that*

$$a_I^{(j)} = c_j a_K^{(j)} + d_j \quad \text{in } \mathbb{Q},$$

with  $c_0 = 2^{-v}$ ,  $d_0 = -2^{-v}$ , and the update rule

$$c_{j+1} = c_j \cdot 2^{v_K^{(j)} - v_I^{(j)}}, \quad d_{j+1} = \frac{3d_j + c_j - 1}{2^{v_I^{(j)}}}.$$

*Proof.* The base case is immediate. Assume the affine relation holds at index  $j$ . Writing  $A_K^{(j)} := 3a_K^{(j)} - 1$ ,  $A_I^{(j)} := 3a_I^{(j)} - 1$ , and  $X_j := 3d_j + c_j$ ,

$$A_I^{(j)} = 3(c_j a_K^{(j)} + d_j) - 1 = c_j(3a_K^{(j)} - 1) + (3d_j + c_j - 1) = c_j A_K^{(j)} + (X_j - 1).$$

Dividing by  $2^{v_I^{(j)}}$  and using  $A_K^{(j)} = 2^{v_K^{(j)}} a_K^{(j+1)}$ :

$$a_I^{(j+1)} = c_j \cdot 2^{v_K^{(j)} - v_I^{(j)}} a_K^{(j+1)} + \frac{X_j - 1}{2^{v_I^{(j)}}},$$

which is the desired affine relation at index  $j + 1$ .  $\square$

We define the  $X$ -invariant of the parallel reduction at  $(r, L)$  as

$$X_{\text{end}}(r, L) := X_J = 3d_J + c_J.$$

This is not invariant under the update step; the name reflects its role as the obstruction certificate (Definition 2.2).

### 2.4. Definition of obstruction residues.

**Definition 2.2** (Obstruction residue). Let  $r$  be a positive odd integer with  $r \neq 1$ , and write  $v := v_2(r - 1) \geq 1$ . We call  $r$  an *obstruction residue at level*  $L > v$  if  $X_{\text{end}}(r, L) = 1$ . We write

$$\mathcal{O}_L := \left\{ r \in \{1, \dots, 2^L\} \mid \begin{array}{l} r \text{ odd, } v_2(r - 1) \geq 1, \\ L > v_2(r - 1), X_{\text{end}}(r, L) = 1 \end{array} \right\}.$$

The definition is algorithmic: given  $r$  and  $L$ , one runs the parallel reduction, observes the terminal pair  $(c_J, d_J)$ , and checks  $3d_J + c_J = 1$ . The computation terminates in at most  $L$  steps.

*Remark 2.3.* A direct enumeration confirms  $\mathcal{O}_L = \emptyset$  for  $1 \leq L \leq 4$ ; the smallest level at which an obstruction residue exists is  $L = 5$ , with  $\mathcal{O}_5 = \{27\}$ . At the next level,  $\mathcal{O}_6 = \{19, 27, 59\}$ , comprising the new atomic residue 19 and the two lifts  $\{27, 59\}$  of  $27 \in \mathcal{O}_5$ .

The connection between this algebraic definition and the dynamics of  $T_-$  trajectories is, briefly, that  $X_{\text{end}} = 1$  implies a synchronization of stopping times: starting from  $r$  and from any  $n \equiv r \pmod{2^L}$ , the stopping-time difference between the  $T_-$  and  $T_-^{(3)}$  iterates — where  $T_-^{(3)}$  denotes the 3-applications-of- $T_-$  variant — remains constant. The present paper develops only the algebraic side; the dynamical interpretation is not used in any proof below.

**2.5. The parity lemma.** The central algebraic fact of this paper is the following.

**Lemma 2.4** (Parity lemma). *Suppose the parallel reduction at  $(r, L)$  terminates with  $X_{\text{end}}(r, L) = 1$ , and write  $v := v_2(r - 1)$ . Then*

$$\delta_J := V_K^{(J)} - V_I^{(J)} - v$$

*is an even integer (possibly negative) with  $\delta_J \in [-(L - 1), L - 1 - v]$ .*

*Proof.* Iterating the update rule from Lemma 2.1 with  $c_0 = 2^{-v}$ :

$$c_j = 2^{-v} \cdot \prod_{i < j} 2^{v_K^{(i)} - v_I^{(i)}} = 2^{V_K^{(j)} - V_I^{(j)} - v} = 2^{\delta_j}.$$

From  $X_J = 3d_J + c_J = 1$ ,  $d_J = (1 - c_J)/3 = (1 - 2^{\delta_J})/3$ . The affine relation at index  $J$  becomes

$$a_I^{(J)} = 2^{\delta_J} a_K^{(J)} + \frac{1 - 2^{\delta_J}}{3} \quad \text{in } \mathbb{Q}.$$

Multiplying by 3:

$$3a_I^{(J)} = 3 \cdot 2^{\delta_J} a_K^{(J)} + 1 - 2^{\delta_J}. \quad (1)$$

We split by sign of  $\delta_J$ . Write  $\delta := \delta_J$  throughout.

*Case  $\delta \geq 0$ .* All terms in (1) are integers. Reducing modulo 3:  $3a_I^{(J)} \equiv 0$  and  $3 \cdot 2^{\delta} a_K^{(J)} \equiv 0$ , so  $1 - 2^{\delta} \equiv 0 \pmod{3}$ , i.e.  $2^{\delta} \equiv 1 \pmod{3}$ . Since  $2 \equiv -1 \pmod{3}$ , this forces  $\delta$  even.

*Case  $\delta < 0$ .* Write  $\delta = -e$ ,  $e > 0$ . Multiplying (1) by  $2^e$ :

$$3 \cdot 2^e a_I^{(J)} = 3 a_K^{(J)} + 2^e - 1.$$

All terms are integers; reducing modulo 3 yields  $2^e - 1 \equiv 0 \pmod{3}$ , hence  $e$  even, hence  $\delta$  even.

In both cases  $\delta$  is even. For the bound, each firing step satisfies  $v_K^{(j)} < L - V_K^{(j)}$  and  $v_I^{(j)} < L - v - V_I^{(j)}$  (else the respective stop would have fired at  $j$ ), so  $V_K^{(J)} \leq L - 1$  and  $V_I^{(J)} \leq L - 1 - v$ ; hence  $\delta_J = V_K^{(J)} - V_I^{(J)} - v \in [-(L - 1), L - 1 - v]$ .  $\square$

**2.6. The shift- $a$  normal form.** Writing  $\delta_J = 2a$  with  $a \in \mathbb{Z}$ , the parity lemma yields

$$c_J = 4^a, \quad d_J = \frac{1 - 4^a}{3}.$$

We call this the *shift- $a$  normal form* and  $a$  the *shift index* of  $r$  at level  $L$ , denoted  $a(r, L)$ . In particular,  $V_K^{(J)} - V_I^{(J)} = 2a + v$ . The obstruction family partitions:

$$\mathcal{O}_L = \bigsqcup_{a \in \mathbb{Z}} \mathcal{O}_L^{G_a}, \quad \mathcal{O}_L^{G_a} := \{r \in \mathcal{O}_L : a(r, L) = a\}.$$

For  $a = 0$  we have  $(c_J, d_J) = (1, 0)$ , meaning  $a_I^{(J)} = a_K^{(J)}$ . By the bound  $\delta_J \in [-(L-1), L-1-v]$  from Lemma 2.4,  $a = \delta_J/2 \in [-(L-1)/2, (L-1-v)/2]$ ; in particular  $|a| \leq \lfloor (L-1)/2 \rfloor$  (using that  $a$  is an integer).

### 2.7. Simultaneous stop and stop types.

**Corollary 2.5** (Simultaneous stop). *If  $X_{\text{end}}(r, L) = 1$ , then the  $K$ -track and  $I$ -track satisfy the termination condition  $v_\alpha \geq v_2(\beta)$  at the same index  $J$ . Moreover, the terminal step valuations satisfy  $v_I^{(J)} = v_K^{(J)} + 2a$ , where  $a$  is the shift index of  $r$  at level  $L$  from the shift- $a$  normal form above.*

*Proof.* From the shift- $a$  normal form,  $a_I^{(J)} = 4^a a_K^{(J)} + (1 - 4^a)/3$ , so  $A_I^{(J)} = 4^a A_K^{(J)}$  and hence  $v_I^{(J)} = v_K^{(J)} + 2a$ . Combining with  $v_2(b_I^{(J)}) = (L-v) - V_I^{(J)}$  and  $v_2(b_K^{(J)}) = L - V_K^{(J)}$ , and using  $V_K - V_I = 2a + v$ :

$$\begin{aligned} v_I^{(J)} \geq v_2(b_I^{(J)}) &\iff v_K^{(J)} + 2a \geq L - v - V_I^{(J)} \\ &\iff v_K^{(J)} \geq L - V_K^{(J)} \\ &\iff v_K^{(J)} \geq v_2(b_K^{(J)}). \quad \square \end{aligned}$$

**Definition 2.6** (Stop type). The *stop type* at  $(r, L)$  is *EE* (equality) if  $v_K^{(J)} = v_2(b_K^{(J)})$  and  $v_I^{(J)} = v_2(b_I^{(J)})$ , and *SS* (strict) if  $v_K^{(J)} > v_2(b_K^{(J)})$  and  $v_I^{(J)} > v_2(b_I^{(J)})$ . By Corollary 2.5 these two cases are exhaustive: a  $K$ -track equality forces an  $I$ -track equality, and a  $K$ -track strict inequality forces an  $I$ -track strict inequality.

**2.8. The parameter  $v$  and the typical case  $v = 1$ .** The constructions above are stated for general  $v := v_2(r-1) \geq 1$ . The typical case is  $v = 1$  (equivalently  $r \equiv 3 \pmod{4}$ ); for  $r \equiv 1 \pmod{4}$  the value of  $v$  is  $\geq 2$ . In every formula above, setting  $v = 1$  recovers the original initial pair  $(c_0, d_0) = (\frac{1}{2}, -\frac{1}{2})$  and the parity relation  $V_K^{(J)} - V_I^{(J)} \equiv 1 \pmod{2}$ . The shift- $a$  normal form  $(c_J, d_J) = (4^a, (1 - 4^a)/3)$  is independent of  $v$ . The subsequent sections work with the general case throughout; explicit calculations use  $v = 1$  wherever this aids the reader.

The  $v \geq 2$  residues contribute a small but non-negligible share of  $\mathcal{O}_L$ : direct enumeration gives, for  $L = 16$ ,  $|\mathcal{O}_L| = 7556$  in total, of which 6130 have  $v = 1$  and the remaining 1426 (roughly 19%) have  $v \geq 2$ . The numbers in the appendix table count all  $v$ .

## 3. THE LIFT THEOREM

We now prove Theorem 1.1: for every  $r_0 \in \mathcal{O}_{L_0}$ , both lifts  $r_+ := r_0$  and  $r_- := r_0 + 2^{L_0}$  are obstructions at level  $L_0 + 1$ . The two halves are stated and proved separately as Theorems 3.1 and 3.2; together they constitute Theorem 1.1. Throughout,  $a \in \mathbb{Z}$  is the shift index of  $r_0$  at level  $L_0$ , and we use the synchronization  $v_I^{(J)} = v_K^{(J)} + 2a$  (Corollary 2.5) freely.



### 3.1. The $r_+$ lift.

**Theorem 3.1.** *Let  $r_0 \in \mathcal{O}_{L_0}^{G_a}$ . Then  $r_+ := r_0 \in \mathcal{O}_{L_0+1}$ .*

*Proof.* Set  $v := v_2(r_0 - 1) \geq 1$ . The starting pairs at  $(r_+, L_0 + 1)$  are  $(r_0, 2^{L_0+1})$  and  $(m_0, 2^{L_0+1-v})$  where  $m_0 = (r_0 - 1)/2^v$ . These differ from the corresponding pairs at  $(r_0, L_0)$  only in the modulus, which is doubled. Since each basic step depends only on  $a_K^{(j)}, a_I^{(j)}$  and not on the modulus (as long as the latter does not constrain the step), the parallel reduction at  $(r_+, L_0 + 1)$  produces identical  $v_K^{(j)}, v_I^{(j)}, c_j, d_j$  for  $j \leq J$  to those at  $(r_0, L_0)$ .

*Stop type EE at  $L_0$ .* Here  $v_K^{(J)} = L_0 - V_K^{(J)}$  exactly. At level  $L_0 + 1$ , the K-track modulus at index  $J$  has 2-adic valuation  $L_0 + 1 - V_K^{(J)} = v_K^{(J)} + 1 > v_K^{(J)}$ , so the step is permitted. Taking one more step, with  $c_J = 4^a, d_J = (1 - 4^a)/3$ :

$$\begin{aligned} c_{J+1} &= 4^a \cdot 2^{v_K^{(J)} - v_I^{(J)}} = 4^a \cdot 2^{-2a} = 1, \\ d_{J+1} &= \frac{3(1 - 4^a)/3 + 4^a - 1}{2^{v_I^{(J)}}} = 0. \end{aligned}$$

After this step,  $V_K^{(J+1)} = L_0$ , so the K-track modulus has valuation 1, and the next step would require  $v_K^{(J+1)} < 1$ . Since  $a_K^{(J+1)}$  is odd,  $3a_K^{(J+1)} - 1$  is even, hence  $v_K^{(J+1)} \geq 1$  — so the next step is impossible. Termination occurs at index  $J + 1$  with  $(c_{J+1}, d_{J+1}) = (1, 0)$ , hence  $X_{\text{end}} = 1$  and  $r_+ \in \mathcal{O}_{L_0+1}^{G_0}$ .

*Stop type SS at  $L_0$ .* Here  $v_K^{(J)} > L_0 - V_K^{(J)}$ , i.e.  $v_K^{(J)} \geq L_0 + 1 - V_K^{(J)}$  (integer), which is exactly the termination condition at level  $L_0 + 1$ . Termination at  $(r_+, L_0 + 1)$  occurs at index  $J$  with terminal data unchanged, so  $r_+ \in \mathcal{O}_{L_0+1}^{G_a}$ .  $\square$

### 3.2. The $r_-$ lift.

**Theorem 3.2.** *Let  $r_0 \in \mathcal{O}_{L_0}^{G_a}$ . Then  $r_- := r_0 + 2^{L_0} \in \mathcal{O}_{L_0+1}$ .*

*Proof.* Write  $a_{K,+}^{(j)}, a_{I,+}^{(j)}$  for the values at  $(r_+, L_0 + 1)$  (from Theorem 3.1) and  $a_{K,-}^{(j)}, a_{I,-}^{(j)}$  for those at  $(r_-, L_0 + 1)$ . The discrepancies

$$\Delta_K^{(j)} := a_{K,-}^{(j)} - a_{K,+}^{(j)}, \quad \Delta_I^{(j)} := a_{I,-}^{(j)} - a_{I,+}^{(j)}$$

satisfy  $\Delta_K^{(0)} = 2^{L_0}, \Delta_I^{(0)} = 2^{L_0-v}$  (the I-track shift  $\Delta_I^{(0)} = (r_- - 1)/2^v - m_0 = 2^{L_0}/2^v$  uses  $v_2(r_- - 1) = \min(v_2(r_0 - 1), L_0) = v$ , valid for  $L_0 > v$ ). We claim that as long as  $V_K^{(j)} < L_0$  and  $V_I^{(j)} < L_0 - v$ , the two reductions take identical local steps,

$$v_2(3a_{K,-}^{(j)} - 1) = v_2(3a_{K,+}^{(j)} - 1) = v_K^{(j)} \quad (2)$$

(and analogously for the I-track), and the discrepancies have the closed form

$$\Delta_K^{(j)} = 2^{L_0 - V_K^{(j)}} \cdot 3^j, \quad \Delta_I^{(j)} = 2^{L_0 - v - V_I^{(j)}} \cdot 3^j. \quad (3)$$

We prove (2) and (3) jointly by induction on  $j$  in the range  $0 \leq j < J$ , where  $J$  is the termination index of the reduction at  $(r_0, L_0)$ . The base case  $j = 0$  is immediate. For the inductive step, assume (3) at step  $j$  and  $j < J$ . Since the K-track at  $(r_0, L_0)$  has not yet terminated,  $v_K^{(j)} < L_0 - V_K^{(j)} = v_2(\Delta_K^{(j)})$ . The decomposition  $3a_{K,-}^{(j)} - 1 = (3a_{K,+}^{(j)} - 1) + 3\Delta_K^{(j)}$  then has its two summands at strictly different valuations  $v_K^{(j)}$  and  $L_0 - V_K^{(j)}$ , so by the ultrametric identity  $v_2(x + y) = \min(v_2x, v_2y)$  when  $v_2x \neq v_2y$ , the sum has valuation exactly  $v_K^{(j)}$ . This proves (2) at step  $j$ . The basic step then yields  $\Delta_K^{(j+1)} = (3\Delta_K^{(j)})/2^{v_K^{(j)}} = 2^{L_0 - V_K^{(j+1)}} \cdot 3^{j+1}$ ; the same calculation on the I-track completes (3) at step  $j + 1$ .

*Stop type SS at  $L_0$ .* At index  $J$  of  $(r_+, L_0 + 1)$ ,  $v_K^{(J)} > L_0 - V_K^{(J)}$ , hence  $v_2(\Delta_K^{(J)}) = L_0 - V_K^{(J)} < v_K^{(J)}$ . The relation  $3a_{K,-}^{(J)} - 1 = (3a_{K,+}^{(J)} - 1) + 3\Delta_K^{(J)}$  now has the two summands at distinct valuations  $v_K^{(J)}$  and  $L_0 - V_K^{(J)}$ , so by the ultrametric inequality the sum has valuation  $v_K^{(J),-} := L_0 - V_K^{(J)} < L_0 + 1 - V_K^{(J)}$ . The K-step at  $(r_-, L_0 + 1)$  at index  $J$  is therefore permitted. The same calculation on the I-track yields  $v_I^{(J),-} := L_0 - v - V_I^{(J)}$ , also permitted, and one more step follows. The required strict inequality  $v_2(\Delta_I^{(J)}) = L_0 - v - V_I^{(J)} < v_I^{(J)}$  is the I-track version of SS: by the synchronisation  $v_I^{(J)} = v_K^{(J)} + 2a$  of Corollary 2.5 together with  $V_K^{(J)} - V_I^{(J)} = 2a + v$ , the K-track strict inequality  $v_K^{(J)} > L_0 - V_K^{(J)}$  is equivalent to  $v_I^{(J)} > L_0 - v - V_I^{(J)}$ . Using  $V_K - V_I = 2a + v$ , the new exponent in the update rule is  $v_K^{(J),-} - v_I^{(J),-} = -2a$ , so

$$c_{J+1} = c_J \cdot 2^{-2a} = 4^a \cdot 4^{-a} = 1, \quad d_{J+1} = \frac{3 \cdot (1 - 4^a)/3 + 4^a - 1}{2^{v_I^{(J),-}}} = 0.$$

After this step  $V_K^{(J+1)} = L_0$  and  $V_I^{(J+1)} = L_0 - v$ , so both moduli have valuation 1 and the next step on either track is impossible for odd  $a_K, a_I$ . Termination occurs at index  $J + 1$  with  $(c_{J+1}, d_{J+1}) = (1, 0)$ , so  $r_- \in \mathcal{O}_{L_0+1}^{G_0}$ .

*Stop type EE at  $L_0$ .* Here  $v_K^{(J)} = L_0 - V_K^{(J)} = v_2(\Delta_K^{(J)})$ , so the two summands of  $3a_{K,-}^{(J)} - 1 = (3a_{K,+}^{(J)} - 1) + 3\Delta_K^{(J)}$  have *equal* valuation  $v_K^{(J)}$ . The ultrametric inequality then gives only  $v_2(A_{K,-}^{(J)}) \geq v_K^{(J)} + 1$ ; that is,  $v_2(A_{K,-}^{(J)}) \geq L_0 + 1 - V_K^{(J)}$ , which is the K-track stop condition at level  $L_0 + 1$ . The same argument on the I-track gives  $v_2(A_{I,-}^{(J)}) \geq L_0 + 1 - v - V_I^{(J)}$ , the I-track stop condition at level  $L_0 + 1$ . Termination therefore occurs at index  $J$  with terminal data unchanged, so  $r_- \in \mathcal{O}_{L_0+1}^{G_a}$ .  $\square$

### 3.3. The lift symmetry.

**Proposition 3.3** (Lift symmetry). *Let  $r_0 \in \mathcal{O}_{L_0}^{G_a}$ . The shift indices of the two level- $(L_0 + 1)$  lifts  $r_+ := r_0$  and  $r_- := r_0 + 2^{L_0}$  depend only on the stop type of  $r_0$  at  $L_0$  (Definition 2.6) as follows (both lifts at level  $L_0 + 1$ ):*

| Stop type of $r_0$ at $L_0$ | shift index of $r_+$ | shift index of $r_-$ |
|-----------------------------|----------------------|----------------------|
| $EE$                        | 0                    | $a$                  |
| $SS$                        | $a$                  | 0                    |

*Proof.* The four entries are read off directly from the case-analysis proofs of Theorems 3.1 and 3.2: each case computes the terminal data of the lift and identifies the resulting shift index.  $\square$

So lifting an obstruction with shift index  $a \neq 0$  always produces one shift- $a$  obstruction and one shift-zero obstruction at the next level; lifting a shift-zero obstruction produces two shift-zero obstructions. This is the symmetry that drives the recurrence developed in Section 6.

#### 4. POSITIVE DENSITY

*Proof of Theorem 1.2.* By Theorem 1.1, every  $r_0 \in \mathcal{O}_{L_0}$  contributes two distinct elements  $r_0$  and  $r_0 + 2^{L_0}$  to  $\mathcal{O}_{L_0+1}$ . Hence  $|\mathcal{O}_{L_0+1}| \geq 2|\mathcal{O}_{L_0}|$ , equivalently  $|\mathcal{O}_{L_0+1}|/2^{L_0+1} \geq |\mathcal{O}_{L_0}|/2^{L_0}$ : the density ratio is non-decreasing. Since  $\mathcal{O}_L \subseteq \{1, \dots, 2^L\}$  the ratio is bounded above by 1. By monotone convergence,  $c_W := \lim_{L \rightarrow \infty} |\mathcal{O}_L|/2^L$  exists in  $[\rho_{L_0}, 1]$  for every  $L_0 \geq 5$ . Substituting the direct enumeration  $|\mathcal{O}_{16}| = 7556$  (Appendix A, Table 3) yields  $c_W \geq 7556/65536 \approx 0.115$ ; the conservative bound  $c_W \geq 3/64$  is recovered at  $L_0 = 6$  from  $\mathcal{O}_6 = \{19, 27, 59\}$  (Section 2).  $\square$

The conservative bound  $\rho_6 = 3/64$  uses only the three explicit obstructions  $\mathcal{O}_6 = \{19, 27, 59\}$ . By raising  $L_0$  and computing  $|\mathcal{O}_{L_0}|$  directly, tighter rigorous bounds for  $c_W$  follow (Table 1):

| $L_0$ | $ \mathcal{O}_{L_0} $ | $\rho_{L_0}$    | rigorous lower bound for $c_W$ |
|-------|-----------------------|-----------------|--------------------------------|
| 6     | 3                     | $\approx 0.047$ | $3/64$                         |
| 10    | 91                    | $\approx 0.089$ | $91/1024$                      |
| 14    | 1776                  | $\approx 0.108$ | $1776/16384$                   |
| 16    | 7556                  | $\approx 0.115$ | $7556/65536$                   |

TABLE 1. Rigorous lower bounds for  $c_W$  obtained from finite-level enumeration of  $\mathcal{O}_{L_0}$ . The ratio  $\rho_{L_0} := |\mathcal{O}_{L_0}|/2^{L_0}$  is non-decreasing in  $L_0$  (Theorem 1.2).

These are not asymptotic estimates: they are finite-level computations producing rigorous bounds at the cost of  $O(L_0 \cdot 2^{L_0})$  operations to enumerate obstructions modulo  $2^{L_0}$ . The exact value of  $c_W$  is not pinned down by this approach; we return to it in Section 6.

## 5. FULL FACTOR COMPLEXITY

**5.1. Setup.** To each  $r \in \mathcal{O}_L$  we associate its binary representation as an  $L$ -bit word, least significant bit first:

$$w_r \in \{0, 1\}^L, \quad w_r[k] := \lfloor r/2^k \rfloor \bmod 2, \quad k = 0, \dots, L-1.$$

In particular  $w_r[0] = 1$  (since  $r$  is odd) and  $w_r[L-1]$  is the leading bit of  $r$  in this  $L$ -bit representation. The *language of obstruction words* at level  $L$  is  $\mathcal{F}_L := \{w_r : r \in \mathcal{O}_L\}$ , and the *factor language* of the obstruction family is

$$\mathcal{F}(\mathcal{O}) := \{u \in \{0, 1\}^* : u \text{ is a factor of } w_r \text{ for some } r \in \mathcal{O}_L, L \geq 5\}.$$

The *factor complexity function* is  $p_W(n) := |\mathcal{F}(\mathcal{O}) \cap \{0, 1\}^n|$ . Trivially  $p_W(n) \leq 2^n$ .

**5.2. The main theorem.**

**Theorem 5.1** (restating Theorem 1.3). *For every  $n \geq 0$ ,  $p_W(n) = 2^n$ .*

*Proof.* The case  $n = 0$  is the empty word, which is trivially a factor of every  $w_r$ , so  $p_W(0) = 1 = 2^0$ . Fix  $n \geq 1$  and  $u = (u_0, u_1, \dots, u_{n-1}) \in \{0, 1\}^n$ . Choose any anchor  $r_0 \in \mathcal{O}_6 = \{19, 27, 59\}$  (verified by direct enumeration; see Appendix A). In fact each of the three anchors individually realises every pattern, as direct enumeration via the lift-symmetry table confirms; for the proof one anchor suffices. Define inductively

$$r_k := r_{k-1} + u_{k-1} \cdot 2^{5+k}, \quad k = 1, 2, \dots, n.$$

We claim  $r_k \in \mathcal{O}_{6+k}$  for each  $k$ . The base  $k = 0$  is by choice of  $r_0$ . For the inductive step from  $k-1$  to  $k$ : if  $u_{k-1} = 0$  then  $r_k = r_{k-1}$ , so  $r_k$  is the  $r_+$  lift and lies in  $\mathcal{O}_{6+k}$  by Theorem 3.1; if  $u_{k-1} = 1$  then  $r_k = r_{k-1} + 2^{5+k}$  is the  $r_-$  lift and lies in  $\mathcal{O}_{6+k}$  by Theorem 3.2.

By construction,  $w_{r_n}[5+k] = u_{k-1}$  for  $k = 1, \dots, n$ , so  $u$  appears as a factor of  $w_{r_n}$  at positions  $6, \dots, 5+n$ . Thus  $u \in \mathcal{F}(\mathcal{O})$  for every  $u \in \{0, 1\}^n$ , hence  $p_W(n) \geq 2^n$ . Combined with the trivial upper bound,  $p_W(n) = 2^n$ .  $\square$

**5.3. Entropy and substitution-shift rigidity.** The *topological entropy* of a language  $\mathcal{L} \subseteq \{0, 1\}^*$  is  $h(\mathcal{L}) := \limsup_{n \rightarrow \infty} (1/n) \log p_{\mathcal{L}}(n)$ ; the factor-complexity function  $p_{\mathcal{L}}$  itself goes back to the foundational symbolic-dynamics papers of Morse–Hedlund [MH38, MH40]. See Lind–Marcus [LM95] for general background on entropy of subshifts and subshifts of finite type, Fogg [Fog02] or Allouche–Shallit [AS03] for factor complexity of substitution and automatic sequences, Cassaigne [Cas97] for the general theory of subword complexity in terms of special factors, and Berthé–Rigo [BR10] and Lothaire [Lot02] for modern collected references on combinatorics on words, automata and number theory.

**Corollary 5.2.** *The topological entropy of  $\mathcal{F}(\mathcal{O})$  is  $\log 2$ .*

This is the maximal value, matching that of the full Bernoulli shift  $\{0, 1\}^{\mathbb{Z}}$ .

**Corollary 5.3.** *The obstruction family  $(\mathcal{O}_L)_{L \geq 5}$ , viewed as a graded family of finite languages over  $\{0, 1\}$ , is*

- (i) *not the language of a proper subshift of finite type, that is, one defined by a non-empty list of forbidden factors;*
- (ii) *not the language of a proper sofic shift in the classical sense (image of a proper SFT under a one-block factor map);*
- (iii) *not the factor language of a primitive substitution shift in the sense of Pansiot [Pan84] or Devyatov [Dev15].*

*Proof.* A non-trivial SFT defined by a non-empty finite list of forbidden subwords has entropy strictly less than  $\log 2$  (Lind–Marcus [LM95]), contradicting Corollary 5.2. A non-trivial sofic shift is the image of an SFT under a one-block factor map and inherits the entropy constraint. Primitive substitution shifts have factor complexity in one of the polynomial classes  $\Theta(1)$ ,  $\Theta(n)$ ,  $\Theta(n \log \log n)$ ,  $\Theta(n \log n)$ ,  $\Theta(n^2)$  (Pansiot [Pan84]; refined by Devyatov [Dev15]), and therefore have topological entropy zero. The obstruction family has  $p_W(n) = 2^n$  and hence entropy  $\log 2 > 0$ , so it cannot be the factor language of a primitive substitution shift.  $\square$

*Remark 5.4.* The restriction to *primitive* substitution shifts in Corollary 5.3(iii) is the scope of the Pansiot–Devyatov classification we invoke; whether  $\mathcal{F}(\mathcal{O})$  can be realised as the factor language of a non-primitive substitution shift remains open (see Section 8).

## 6. THE $G_a$ CLASSIFICATION

**6.1. Atomic obstructions.** An obstruction  $r \in \mathcal{O}_L$  is *atomic at level  $L$*  if  $r \bmod 2^{L-1} \notin \mathcal{O}_{L-1}$ . We write  $\mathcal{A}_L \subseteq \mathcal{O}_L$  and  $\mathcal{A}_L^{G_a} := \mathcal{A}_L \cap \mathcal{O}_L^{G_a}$ .

**Lemma 6.1** (Atom decomposition). *For every  $L \geq 5$ ,*

$$|\mathcal{O}_L| = \sum_{L_0=5}^L |\mathcal{A}_{L_0}| \cdot 2^{L-L_0}.$$

*Proof.* We show that every  $r \in \mathcal{O}_L$  has a unique smallest  $L_0 \geq 5$  with  $r \bmod 2^{L_0} \in \mathcal{A}_{L_0}$ . *Existence:* starting from  $r$  at level  $L$ , consider the descending sequence  $r_k := r \bmod 2^{L-k}$  for  $k = 0, 1, \dots, L-5$ . By the lift theorem (Theorem 1.1), whenever  $r_{k+1} \in \mathcal{O}_{L-k-1}$ , the level- $(L-k)$  residue  $r_k = r \bmod 2^{L-k}$  — being one of the two lifts of  $r_{k+1}$  at level  $L-k$  — lies in  $\mathcal{O}_{L-k}$ ; the converse need not hold. Let  $L_0$  be the smallest level  $\geq 5$  for which  $r \bmod 2^{L_0} \in \mathcal{O}_{L_0}$ . Then  $r \bmod 2^{L_0-1} \notin \mathcal{O}_{L_0-1}$  (either because  $L_0 = 5$  and  $\mathcal{O}_4 = \emptyset$  by Remark 2.3, or by minimality), so  $r \bmod 2^{L_0} \in \mathcal{A}_{L_0}$ . *Uniqueness:* the level  $L_0$  is the smallest such by definition, hence unique. Given the atomic anchor  $r_0 := r \bmod 2^{L_0}$ , the residue  $r$  is determined by  $r_0$  together with the bits  $\{u_{L_0}, \dots, u_{L-1}\}$  encoding the lift choices (Theorems 3.1, 3.2). Each atomic anchor at level  $L_0$  thus generates  $2^{L-L_0}$  obstructions at level  $L$ .  $\square$

## 6.2. No shift-zero atoms.

**Theorem 6.2.** *For every  $L \geq 7$ ,  $|\mathcal{A}_L^{G_0}| = 0$ .*

The conclusion also holds at  $L = 5, 6$  by direct enumeration (see Corollary 6.3); the bound  $L \geq 7$  in the statement is conservative. The case analysis below applies verbatim at  $L = 6$  and is vacuous at  $L = 5$ , where  $|\mathcal{O}_5^{G_0}| = 0$ ; we defer both base cases to Corollary 6.3 for clarity.

*Proof.* Fix  $L \geq 7$  and suppose  $r \in \mathcal{O}_L^{G_0}$ , so  $X_{\text{end}}(r, L) = 1$  with shift index  $a = 0$ , i.e.  $(c_J, d_J) = (1, 0)$ . Write  $r' := r \bmod 2^{L-1}$ . We first note  $J \geq 2$ : at  $J = 0$  the terminal pair is  $(2^{-v}, -2^{-v})$ , whose second coordinate  $-2^{-v}$  is nonzero for every  $v$ , contradicting  $d_J = 0$ ; at  $J = 1$  the update rule gives  $d_1 = (3 \cdot (-2^{-v}) + 2^{-v} - 1)/2^{v_I^{(0)}} = -(1 + 2^{1-v})/2^{v_I^{(0)}}$ , whose numerator is strictly negative for  $v \geq 1$  (since  $2^{1-v} + 1 > 0$ ), so  $d_1 \neq 0$  and again contradicting  $d_J = 0$ . Hence  $J \geq 2$ , so the case distinction below over the indices  $J-1$  and  $J$  is well-defined. The parallel reduction at  $(r', L-1)$  starts from the same residue class  $a_K \bmod 2^{L-1}, a_I \bmod 2^{L-1-v}$  as at  $(r, L)$ , with the modulus reduced by one power of two; the two K-tracks (resp. I-tracks) differ as integers by  $\Delta_K^{(j)} = 2^{L-1-V_K^{(j)}} \cdot 3^j$  (resp.  $\Delta_I^{(j)} = 2^{L-1-v-V_I^{(j)}} \cdot 3^j$ ), in direct analogy with the lift-minus calculation of Theorem 3.2. Whenever  $v_K^{(j)} < L-1-V_K^{(j)}$ , the ultrametric identity forces  $v_2(3a_K^{(j)} - 1)$  to agree at  $(r, L)$  and  $(r', L-1)$ , so the per-step valuations  $v_K^{(j)}, v_I^{(j)}$  and hence the affine data  $(c_j, d_j)$  coincide between the two reductions until the level- $(L-1)$  stop fires.

Termination of the reduction at  $(r, L)$  at index  $J$  means at least one track satisfies its stop condition there; by Corollary 2.5, both do. Since  $a = 0$ , at the terminal index  $J$  the synchronisation  $v_I^{(J)} = v_K^{(J)} + 2a = v_K^{(J)}$  and  $V_K^{(J)} - V_I^{(J)} = v$  holds (Lemma 2.4), so the K-track stop condition  $v_K^{(J)} \geq L - V_K^{(J)}$  and the I-track stop condition  $v_I^{(J)} \geq (L - v) - V_I^{(J)}$  at  $j = J$  are equivalent: substituting  $V_I^{(J)} = V_K^{(J)} - v$  and  $v_I^{(J)} = v_K^{(J)}$  into the I-track condition reduces it to  $v_K^{(J)} \geq L - V_K^{(J)}$ , so both stop conditions at  $j = J$  read  $V_K^{(J)} + v_K^{(J)} \geq L$ . (The K- and I-track stop conditions are not in general equivalent at  $j < J$ , so we rule out child stops on each track separately below; the equivalence is used only at the terminal index  $J$ , where it follows from  $a = 0$ .) We split by the K-track value  $V_K^{(J)}$  (equivalently  $V_K^{(J-1)} + v_K^{(J-1)}$ ): either  $V_K^{(J)} < L-1$  (Case A) or  $V_K^{(J)} \geq L-1$  (Case B); these exhaust the integer possibilities. We first rule out the K-track stop at level  $L-1$  firing at any earlier index  $J' \leq J-2$ : in that case  $V_K^{(J')} + v_K^{(J')} \geq L-1$ , the  $(r, L)$  reduction takes its  $J'$ -th step (the level- $L$  stop has not yet fired since  $J' < J$ ), giving  $V_K^{(J'+1)} \geq L-1$ , so the K-track modulus at level  $L$  at index  $J'+1$  has valuation  $L - V_K^{(J'+1)} \leq 1$ . Since  $a_K^{(J'+1)}$  is odd,  $v_K^{(J'+1)} \geq 1$ , so the next step is impossible: the  $(r, L)$  reduction would stop at  $J'+1 \leq J-1$ , contradicting

the definition of  $J$ . The same argument applied to the I-track rules out the child I-track stop firing at any index  $J' \leq J - 2$ : an I-stop at  $J'$  would force  $V_I^{(J'+1)} \geq L - 1 - v$ , so the I-track modulus at level  $L$  at index  $J' + 1$  would have valuation  $L - v - V_I^{(J'+1)} \leq 1$  and  $v_I^{(J'+1)} \geq 1$  (since  $a_I^{(J'+1)}$  is odd) would terminate the parent at  $J' + 1 \leq J - 1$ , again contradicting the definition of  $J$ . At index  $J - 1$ , a child I-stop requires  $V_I^{(J)} \geq L - 1 - v$ ; by the terminal synchronisation  $V_I^{(J)} = V_K^{(J)} - v$  (from  $a = 0$ , equivalently  $\delta_J = 0$ ), this is exactly  $V_K^{(J)} \geq L - 1$  — the Case B branch below — so in Case A neither child stop fires before index  $J$ .

*Case A:*  $V_K^{(J)} < L - 1$ . The K-track stop at level  $L - 1$  does not fire at any index  $\leq J - 1$  (the indices  $J' \leq J - 2$  are ruled out by the preceding paragraph; the index  $J' = J - 1$  would force  $V_K^{(J)} = V_K^{(J-1)} + v_K^{(J-1)} \geq L - 1$ , contradicting the Case-A hypothesis  $V_K^{(J)} < L - 1$ ), so the  $(r', L - 1)$  reduction runs at least as far as index  $J$ ; at  $j = J$  we have  $V_K^{(J)} + v_K^{(J)} \geq L > L - 1$ , hence the K-track stop at level  $L - 1$  fires at  $J$  and the reduction terminates there. (The I-track stop at level  $L - 1$  also fires at  $J$  by the same chain — at  $j = J$  the synchronisation  $v_I^{(J)} = v_K^{(J)}$ ,  $V_K^{(J)} - V_I^{(J)} = v$  holds — but only the first track to stop matters for termination.) The terminal data of the two reductions coincide:  $(c_J, d_J) = (1, 0)$ , so  $r' \in \mathcal{O}_{L-1}^{G_0}$  and  $r$  is not atomic.

*Case B:*  $V_K^{(J)} \geq L - 1$ , equivalently  $V_K^{(J-1)} + v_K^{(J-1)} \geq L - 1$ . Combined with the standing  $V_K^{(J-1)} + v_K^{(J-1)} < L$  (since the level- $L$  stop does not fire at  $J - 1$ ), the only integer value is  $V_K^{(J)} = L - 1$ , so the K-track stop at level  $L - 1$  fires at  $J - 1$  and the  $(r', L - 1)$  reduction terminates one step earlier, at index  $J - 1$ , with terminal data  $(c_{J-1}, d_{J-1})$  (whether the I-track stop at level  $L - 1$  also fires at  $J - 1$  is immaterial for termination). From the update rule (Lemma 2.1) at index  $J - 1$ ,

$$c_J = c_{J-1} \cdot 2^{v_K^{(J-1)} - v_I^{(J-1)}}, \quad d_J = \frac{3d_{J-1} + c_{J-1} - 1}{2^{v_I^{(J-1)}}}.$$

Setting  $c_J = 1$ ,  $d_J = 0$  and reading off the numerator of  $d_J$ :  $3d_{J-1} + c_{J-1} - 1 = 0$ , i.e.  $X_{J-1} = 1$ . By the parity lemma applied at the terminal index  $J - 1$  of the reduction at  $(r', L - 1)$ , the terminal pair is in shift- $a'$  normal form for some  $a' \in \mathbb{Z}$ , so  $r' \in \mathcal{O}_{L-1}^{G_{a'}}$ .

In either case,  $r' \in \mathcal{O}_{L-1}$ , so  $r$  is not atomic.  $\square$

**Corollary 6.3.** *For every  $L \geq 5$ ,  $|\mathcal{A}_L^{G_0}| = 0$ .*

*Proof.* For  $L \geq 7$ , this is Theorem 6.2. At the two base levels direct enumeration (verified by `count_obstructions.py`) gives  $\mathcal{A}_5 = \{27\}$  and  $\mathcal{A}_6 = \{19\}$ , each with shift index  $a = 1 \neq 0$ , so  $|\mathcal{A}_5^{G_0}| = |\mathcal{A}_6^{G_0}| = 0$  as well.  $\square$

**6.3. Strict lift balance and series representation.** Write  $g_L := |\mathcal{O}_L^{G_0}|$  and  $h_L := |\mathcal{O}_L^{G_{\neq 0}}|$ . The lift symmetry of Proposition 3.3 gives, per parent at  $L - 1$  (Table 2):

| Parent at $L - 1$ | Contribution to $g_L$ | Contribution to $h_L$ |
|-------------------|-----------------------|-----------------------|
| $G_0$             | 2                     | 0                     |
| $G_a, a \neq 0$   | 1                     | 1                     |

TABLE 2. Per-parent contribution of a level- $(L - 1)$  obstruction to the level- $L$  shift-zero and shift-nonzero populations under the two lifts  $r_{\pm}$ .

Summing and adding the atomic contributions, which by Corollary 6.3 occur only in  $h_L$ :

**Corollary 6.4** (Strict lift balance). *For every  $L \geq 6$ ,*

$$g_L = 2g_{L-1} + h_{L-1}, \quad h_L = h_{L-1} + |\mathcal{A}_L^{G_{\neq 0}}|.$$

*Adding,  $|\mathcal{O}_L| = 2|\mathcal{O}_{L-1}| + |\mathcal{A}_L^{G_{\neq 0}}|$ .*

**Theorem 6.5.** *The density constant  $c_W$  admits the convergent series representation*

$$c_W = \frac{3}{64} + \sum_{L \geq 7} \frac{|\mathcal{A}_L^{G_{\neq 0}}|}{2^L},$$

*with non-negative terms. In particular,  $|\mathcal{A}_L^{G_{\neq 0}}|/2^L \rightarrow 0$  as  $L \rightarrow \infty$ .*

*Proof.* By Lemma 6.1 every  $r \in \mathcal{O}_L$  has a unique atomic anchor at some level  $L_0 \in \{5, 6, \dots, L\}$ . By Corollary 6.3,  $|\mathcal{A}_{L_0}^{G_0}| = 0$  for every  $L_0 \geq 5$ , so each atom contributes to  $\mathcal{A}_{L_0}^{G_{\neq 0}}$ . Direct enumeration at the two base levels (verified by `count_obstructions.py`) gives  $\mathcal{A}_5 = \{27\}$  and  $\mathcal{A}_6 = \{19\}$ , both with shift index  $a = 1$ , so  $|\mathcal{A}_5^{G_{\neq 0}}| = |\mathcal{A}_6^{G_{\neq 0}}| = 1$ . Hence for every  $L \geq 6$ ,

$$|\mathcal{O}_L| = 2^{L-5} + 2^{L-6} + \sum_{L_0=7}^L |\mathcal{A}_{L_0}^{G_{\neq 0}}| \cdot 2^{L-L_0} = 3 \cdot 2^{L-6} + \sum_{L_0=7}^L |\mathcal{A}_{L_0}^{G_{\neq 0}}| \cdot 2^{L-L_0}.$$

Dividing both sides by  $2^L$ ,

$$\rho_L = \frac{|\mathcal{O}_L|}{2^L} = \frac{3}{64} + \sum_{L_0=7}^L \frac{|\mathcal{A}_{L_0}^{G_{\neq 0}}|}{2^{L_0}}.$$

By Theorem 1.2,  $\rho_L$  is non-decreasing in  $L$  and bounded above by 1, so its limit  $c_W$  exists in  $(0, 1]$ . The right-hand side is the  $L$ -th partial sum of a non-negative series with fixed first term  $3/64$ ; since the partial sums converge to  $c_W$ , the series converges, with sum  $c_W$  and tail  $|\mathcal{A}_L^{G_{\neq 0}}|/2^L \rightarrow 0$  as  $L \rightarrow \infty$ .  $\square$



**6.4. The remaining open question.** Theorem 6.5 reduces the question of  $c_W$ 's exact value to controlling the growth of  $|\mathcal{A}_L^{G \neq 0}|$ . A bound of the form  $|\mathcal{A}_L^{G \neq 0}| \leq C \cdot \alpha^L$  with  $\alpha < 2$  would give a geometrically convergent series with explicit error bounds. We do not establish such a bound here; we return to the question in Section 8.

## 7. THE $T_+$ SIDE

The classical Collatz map  $T_+$  admits the same obstruction theory. We transfer the framework of Section 2 to  $T_+$  and show that the residue involution  $r \mapsto \bar{r} := 2^L - r$  realises an explicit bijection between the two obstruction families, step-by-step at the level of the parallel reduction.

**7.1. The  $T_+$  parallel reduction.** Fix a positive odd integer  $r$  with  $v_2(r+1) \geq 1$ , and write  $v := v_2(r+1)$  and  $m := (r+1)/2^v$ ; then  $m$  is odd. For a level  $L > v$ , run the parallel reduction with  $T_+$ :

- the  $K$ -track, starting from  $(a_K^{(0)}, b_K^{(0)}) := (r, 2^L)$  with basic step  $a'_K = (3a_K + 1)/2^{v_K}$ ,  $v_K := v_2(3a_K + 1)$ ;
- the  $I$ -track, starting from  $(a_I^{(0)}, b_I^{(0)}) := (m, 2^{L-v})$  with the analogous basic step.

The initial relation  $m = (r+1)/2^v$  writes as  $a_I^{(0)} = 2^{-v}a_K^{(0)} + 2^{-v}$ , so the affine relation  $a_I^{(j)} = c_j a_K^{(j)} + d_j$  holds at index 0 with  $c_0 = 2^{-v}$ ,  $d_0 = +2^{-v}$ . The derivation of Lemma 2.1, applied with  $T_+$  in place of  $T_-$  and using  $3a_I + 1 = c_j(3a_K + 1) + (3d_j - c_j + 1)$ , yields the update rule

$$c_{j+1} = c_j \cdot 2^{v_K^{(j)} - v_I^{(j)}}, \quad d_{j+1} = \frac{3d_j - c_j + 1}{2^{v_I^{(j)}}}.$$

(The numerator changes sign relative to the  $T_-$  rule.) We define the  $T_+$ -analogue of the  $X$ -invariant by

$$X_{\text{end}}^+(r, L) := c_J - 3d_J,$$

where  $J$  is the termination index. An odd  $r$  with  $v_2(r+1) \geq 1$  and  $L > v_2(r+1)$  is a  $T_+$ -obstruction residue at level  $L$  if  $X_{\text{end}}^+(r, L) = 1$ ; we write  $\mathcal{O}_L^+ \subseteq \{1, \dots, 2^L\}$  for the set of such residues.

*Remark 7.1.* The sign convention  $X_{\text{end}}^+ := c - 3d$  (rather than  $3d + c$ ) is the one forced by the affine algebra: with this combination the analogue of the parity lemma (Lemma 2.4) yields the shift- $a$  normal form  $(c_J, d_J) = (4^a, (4^a - 1)/3)$ , and the residue-level bijection of Theorem 7.3 below is exact.

## 7.2. The residue involution.

**Lemma 7.2** (Residue-level involution). *Let  $r$  be odd with  $v := v_2(r-1) \geq 1$  and  $L > v$ , and set  $\bar{r} := 2^L - r$ . In the conclusions below, write  $V_K^{(j)} := \sum_{i < j} v_K^{(i)}$  and  $V_I^{(j)} := \sum_{i < j} v_I^{(i)}$  for the cumulative valuations along the two*

tracks; the proof shows that they coincide across the  $T_+$  and  $T_-$  reductions, so a single notation suffices. Conclusions:

- (i)  $\bar{r}$  is odd with  $v_2(\bar{r} + 1) = v < L$  (so the  $T_+$  reduction at  $\bar{r}$  in the sense of Section 7 uses the same parameter  $v$ );
- (ii) the  $T_-$  parallel reduction at  $(r, L)$  and the  $T_+$  parallel reduction at  $(\bar{r}, L)$ , both with parameter  $v$ , terminate at the same index  $J$ ;
- (iii) for every  $0 \leq j \leq J$ ,

$$a_K^{+, (j)} \equiv -a_K^{-, (j)} \pmod{2^{L-V_K^{(j)}}}, \quad a_I^{+, (j)} \equiv -a_I^{-, (j)} \pmod{2^{L-v-V_I^{(j)}}}, \quad (4)$$

and  $v_K^{+, (j)} = v_K^{-, (j)}$ ,  $v_I^{+, (j)} = v_I^{-, (j)}$  whenever  $j < J$ ;

- (iv) the affine data satisfy

$$(c_j^+, d_j^+) = (c_j^-, -d_j^-) \quad (0 \leq j \leq J). \quad (5)$$

*Proof.* (i):  $\bar{r} + 1 = 2^L - (r - 1)$  has, by the ultrametric identity with  $v_2(r - 1) = v < L$ , 2-adic valuation exactly  $v$ . Oddness of  $\bar{r}$  is immediate.

(iii) and (ii) jointly: we prove the residue congruences in (4) for  $j = 0, 1, \dots$  by induction, and deduce simultaneous termination and the valuation equalities along the way.

*Base case* ( $j = 0$ ). K-track:  $a_K^{+, (0)} = \bar{r} = 2^L - r = -a_K^{-, (0)} + 2^L$ . I-track:  $a_I^{-, (0)} = (r - 1)/2^v$  and  $a_I^{+, (0)} = (\bar{r} + 1)/2^v = (2^L - (r - 1))/2^v = 2^{L-v} - a_I^{-, (0)}$ . Both congruences in (4) hold at  $j = 0$ . For the valuation equality at  $j = 0$ , suppose the  $T_-$  reduction is still running, i.e.  $v_K^{-, (0)} < L$ . Then  $3\bar{r} + 1 = 3 \cdot 2^L - (3r - 1)$  has summands of valuations  $L$  and  $v_K^{-, (0)}$  respectively, with  $v_K^{-, (0)} < L$ ; the ultrametric identity gives  $v_K^{+, (0)} = v_K^{-, (0)}$ . The analogous calculation on the I-track gives  $v_I^{+, (0)} = v_I^{-, (0)}$ .

*Inductive step.* Assume (4) at index  $j$  and that both reductions have not yet terminated at index  $j$  (so  $v_K^{(j)} < L - V_K^{(j)}$  and  $v_I^{(j)} < L - v - V_I^{(j)}$ , with valuations equal across  $+/-$  by the inductive valuation equalities). For the K-track,

$$3a_K^{+, (j)} + 1 \equiv 3(-a_K^{-, (j)}) + 1 = -(3a_K^{-, (j)} - 1) \pmod{2^{L-V_K^{(j)}}}.$$

The right-hand side has valuation  $v_K^{(j)} < L - V_K^{(j)}$ , so the left-hand side has the same valuation  $v_K^{+, (j)} = v_K^{(j)}$ , and dividing through by  $2^{v_K^{(j)}}$  gives

$$a_K^{+, (j+1)} = \frac{3a_K^{+, (j)} + 1}{2^{v_K^{(j)}}} \equiv -\frac{3a_K^{-, (j)} - 1}{2^{v_K^{(j)}}} = -a_K^{-, (j+1)} \pmod{2^{L-V_K^{(j+1)}}}.$$

The same calculation on the I-track gives the analogous congruence and  $v_I^{+, (j)} = v_I^{(j)}$ .

*Simultaneous termination.* The termination condition of the  $(\bar{r}, L)$  reduction at index  $j$  is  $v_K^{+, (j)} \geq L - V_K^{(j)}$  or  $v_I^{+, (j)} \geq L - v - V_I^{(j)}$ . We show

termination at exactly  $J$  in two directions. *No earlier stop*: if the  $T_+$  reduction stopped at some  $j' < J$ , then by the inductive step at  $j'$  (valid because both reductions still run at  $j'$ ) the valuation equalities  $v_K^{+, (j')} = v_K^{-, (j')}$  and  $v_I^{+, (j')} = v_I^{-, (j')}$  hold, so the  $T_-$  reduction would also satisfy its stop condition at  $j'$ , contradicting the definition of  $J$ . *Stop at  $J$* : the residue congruence (4) at  $j = J$  follows by applying the inductive step one final time (which is valid because the inductive hypothesis at  $j = J - 1$  only requires the reductions to be running there, not at  $J$ ). Combining this congruence with the  $T_-$  K-track stop  $3a_K^{-, (J)} - 1 \equiv 0 \pmod{2^{L-V_K^{(J)}}}$  yields  $3a_K^{+, (J)} + 1 \equiv -(3a_K^{-, (J)} - 1) \equiv 0 \pmod{2^{L-V_K^{(J)}}}$ , so  $v_K^{+, (J)} \geq L - V_K^{(J)}$  and the  $T_+$  K-track stops at  $J$ ; the analogous calculation on the I-track gives  $v_I^{+, (J)} \geq L - v - V_I^{(J)}$ .

(iv): At  $j = 0$ ,  $c_0^+ = 2^{-v} = c_0^-$  and  $d_0^+ = +2^{-v} = -d_0^-$ . Inductively, given  $(c_j^+, d_j^+) = (c_j^-, -d_j^-)$  and the valuation equalities  $v_K^{+, (j)} = v_K^{-, (j)}$ ,  $v_I^{+, (j)} = v_I^{-, (j)}$ ,

$$\begin{aligned} c_{j+1}^+ &= c_j^+ \cdot 2^{v_K^{(j)} - v_I^{(j)}} = c_{j+1}^-, \\ d_{j+1}^+ &= \frac{3d_j^+ - c_j^+ + 1}{2^{v_I^{(j)}}} = -\frac{3d_j^- + c_j^- - 1}{2^{v_I^{(j)}}} = -d_{j+1}^-. \end{aligned} \quad \square$$

### 7.3. The bijection theorem.

**Theorem 7.3.** *For every level  $L$ , the involution  $r \mapsto \bar{r} := 2^L - r$  restricts to a bijection  $\mathcal{O}_L \leftrightarrow \mathcal{O}_L^+$ . Consequently  $|\mathcal{O}_L^+| = |\mathcal{O}_L|$ , and Theorems 1.1, 1.2, and 1.3 hold verbatim for  $\mathcal{O}^+$ , with the same density constant  $c_W$  and factor complexity function  $p_W^+(n) = 2^n$ .*

*Proof.* For  $r$  odd with  $v_2(r - 1) \geq 1$  and  $L > v_2(r - 1)$ , equation (5) of Lemma 7.2(iv) at the terminal index  $J$  together with the sign convention  $X_{\text{end}}^+ := c - 3d$  of Remark 7.1 gives

$$X_{\text{end}}^+(\bar{r}, L) = c_J^+ - 3d_J^+ = c_J^- - 3 \cdot (-d_J^-) = c_J^- + 3d_J^- = X_{\text{end}}(r, L).$$

Hence  $r \in \mathcal{O}_L$  iff  $\bar{r} \in \mathcal{O}_L^+$ . Since  $r \mapsto \bar{r}$  is an involution on the odd residues in  $\{1, \dots, 2^L - 1\}$ , it restricts to a bijection  $\mathcal{O}_L \leftrightarrow \mathcal{O}_L^+$ . Concretely, at  $L = 5$  this gives  $\mathcal{O}_5^+ = \{5\}$ , and at  $L = 6$  it gives  $\mathcal{O}_6^+ = \{2^6 - r : r \in \mathcal{O}_6\} = \{5, 37, 45\}$ , the anchor set parallel to  $\mathcal{O}_6$  in the  $T_+$  analogue of the factor-complexity construction below.

For the remaining claims, observe that Lemma 7.2 conjugates each  $T_-$  basic step into the corresponding  $T_+$  basic step. The  $T_+$  lift theorem (the analogue of Theorem 1.1) follows by composing the involution with the  $T_-$  lift: given  $\bar{r}_0 \in \mathcal{O}_{L_0}^+$ , set  $r_0 := 2^{L_0} - \bar{r}_0 \in \mathcal{O}_{L_0}$ ; by Theorem 1.1, both  $r_0$  and  $r_0 + 2^{L_0}$  lie in  $\mathcal{O}_{L_0+1}$ , and their level- $(L_0 + 1)$  involution images  $2^{L_0+1} - r_0 = \bar{r}_0 + 2^{L_0}$  and  $2^{L_0+1} - (r_0 + 2^{L_0}) = \bar{r}_0$  lie in  $\mathcal{O}_{L_0+1}^+$  by the level- $(L_0 + 1)$  version of the bijection. Hence both  $T_+$ -lifts of  $\bar{r}_0$  are again  $T_+$ -obstructions. Positive density and full factor complexity for  $\mathcal{O}^+$  then

propagate exactly as in Sections 4 and 5, with  $\mathcal{O}^+$  in place of  $\mathcal{O}$  throughout the argument.  $\square$

## 8. OPEN QUESTIONS

**8.1. The exact value of  $c_W$ .** The principal open question is to determine  $c_W$  exactly, or equivalently to give a quantitative bound on the growth of  $|\mathcal{A}_L^{G \neq 0}|$ . Three approaches present themselves, each requiring substantial additional work.

**(i) Substitution-theoretic.** Stérin [Sté20] realizes  $T_+$  as a four-state finite-state transducer between bases 2 and 3. An analogous construction for the parallel reduction, combined with Pansiot’s classification [Pan84] (refined by Devyatov [Dev15]), would determine the asymptotic growth class of  $\mathcal{A}_L^{G \neq 0}$ . The relevant question concerns *non-primitive* substitution shifts: the primitive case is ruled out by Corollary 5.3, so any realisation must lie outside that classification (Remark 5.4).

**(ii) Markov-operator-spectral.** Wirsching [Wir98] constructs a Markov operator  $W_3$  on residue-density functions for the  $T_+$  predecessor problem. An analogous operator for the parallel reduction would have an action on  $(c, d)$ -space; its spectrum would determine the asymptotic growth of the shift-class populations  $|\mathcal{O}_L^{G_a}|$ . A unified treatment — for instance, expressing  $c_W$  in terms of Wirsching’s invariant density  $\varphi$  — appears not to be in the literature.

**(iii) Generating-function-combinatorial.** A direct enumeration via generating functions in  $L$  is in principle accessible, but the absence of a closed-form recursion — the count depends on the joint distribution of stopping times and shift indices — makes the analysis delicate.

**8.2. Nontrivial cycles.** The shift-zero class  $\mathcal{O}_L^{G_0}$  encodes residues for which the two parallel stopping times remain rigidly tied. Whether the existence (or absence) of additional nontrivial  $T_-$ -cycles is reflected in the structure of  $\mathcal{O}_L^{G_0}$  for large  $L$  — in particular, whether one can extract a Diophantine obstruction from the strict lift balance of Corollary 6.4 — is a natural and apparently open question. Classical lower bounds on the length of nontrivial  $T_+$ -cycles, due to Eliahou [Eli93] and extended by Belaga and Mignotte [BM06], indicate how rigidly the Diophantine geometry constrains cycle structure; an analogous bound in terms of  $\mathcal{O}_L^{G_0}$  would be of independent interest.

## EXTENSION TO THE $(3n + b)$ FAMILY

The constructive lift, the positive-density bound, and the full-factor-complexity result extend to the family  $T_b(n) := (3n + b)/2^{v_2(3n+b)}$  for every odd  $b$ . The bridge is a multiplicative residue-bijection on  $\{1, \dots, 2^L\}$  that conjugates the obstruction structure across the family and preserves the shift index; the involution  $r \mapsto 2^L - r$  used in Section 7 is a special case. A detailed treatment is in preparation as a separate paper.

## FURTHER EXTENSIONS

Significant partial results are also known for general multiplier  $a$  — in particular structure theorems for  $a = -1$  and for the negative-Mersenne family  $a = -(2^v - 1)$  with explicit lower bounds on  $c_W$ . A complete treatment of the  $(ax + b)$  family raises new phenomena (Mersenne asymmetry, a sub-class hierarchy, a signed sync index) and is the subject of ongoing work.

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## APPENDIX A. COMPUTATIONAL VERIFICATION

The algorithmic  $X$ -invariant criterion of Section 2 is implemented in Python in the supporting code that accompanies this paper, collected in the directory `code/` alongside the manuscript source and mirrored at [https://github.com/yaccob/collatz/tree/main/manuscripts/obstruction\\_residues/code](https://github.com/yaccob/collatz/tree/main/manuscripts/obstruction_residues/code). The directory contains the verification scripts for each rigorous result (lift theorem, atom decomposition, the no-shift-zero-atom theorem (Theorem 6.2), full factor complexity, simultaneous stop) together with a `README.md` mapping each script to the result it certifies. For each level  $L \leq L_{\max}$ , the relevant script enumerates all odd residues  $r$  with  $v_2(r - 1) \geq 1$ , runs the parallel reduction, and records the terminal data. The set  $\mathcal{O}_L$  is read off as the preimage of  $\{(c, d) : 3d + c = 1\}$ .

We have run this enumeration through  $L = 16$ , producing the counts of Table 3.

The data satisfies the strict lift balance  $|\mathcal{O}_L| = 2|\mathcal{O}_{L-1}| + |\mathcal{A}_L^{G \neq 0}|$  of Corollary 6.4 at every level, and is consistent with Theorems 1.1, 6.2, and 6.5 (series convergence).

The rigorous bound  $c_W \geq 7556/65536 \approx 0.115$  of Theorem 1.2 is conditional on the correctness of this enumeration. The check is self-validating in a useful sense: the strict lift balance of Corollary 6.4 is asserted row-by-row by the script `count_obstructions.py` against the previous-level count, so any drift in the enumeration would manifest as an immediate hard failure rather than silently shifting the bound. Each rigorous-verification script exits non-zero on any violation of the asserted theorem (a convention checked statically by `meta_test_exit_convention.py`), so the bound is reproducible end-to-end by running the script.

| $L$ | $ \mathcal{O}_L $ | $ \mathcal{O}_L^{G_0} $ | $ \mathcal{O}_L^{G_{\neq 0}} $ | $ \mathcal{A}_L^{G_{\neq 0}} $ |
|-----|-------------------|-------------------------|--------------------------------|--------------------------------|
| 5   | 1                 | 0                       | 1                              | 1                              |
| 6   | 3                 | 1                       | 2                              | 1                              |
| 7   | 8                 | 4                       | 4                              | 2                              |
| 8   | 19                | 12                      | 7                              | 3                              |
| 9   | 42                | 31                      | 11                             | 4                              |
| 10  | 91                | 73                      | 18                             | 7                              |
| 11  | 194               | 164                     | 30                             | 12                             |
| 12  | 409               | 358                     | 51                             | 21                             |
| 13  | 855               | 767                     | 88                             | 37                             |
| 14  | 1776              | 1622                    | 154                            | 66                             |
| 15  | 3671              | 3398                    | 273                            | 119                            |
| 16  | 7556              | 7069                    | 487                            | 214                            |

TABLE 3. Direct enumeration counts at levels  $L = 5, \dots, 16$ : total obstructions, shift-zero and shift-nonzero subclasses, and atomic shift-nonzero obstructions. The decomposition  $|\mathcal{O}_L| = |\mathcal{O}_L^{G_0}| + |\mathcal{O}_L^{G_{\neq 0}}|$  is read off each row, and the strict lift balance  $|\mathcal{O}_L| = 2|\mathcal{O}_{L-1}| + |\mathcal{A}_L^{G_{\neq 0}}|$  (Corollary 6.4) holds row-by-row against the previous one for  $L \geq 6$ .

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